Qadam / ArcSolo

Test scenarios, limitations and privacy — Build with AI, Track 1

Prepared 18 October 2026. Project documentation: English. Pitch: Russian. This report accompanies the working prototype, automated OSM collector, source repository and presentation.

Verification summary

32 engine/API/Worker tests; 160 combinations of location, time, interests and weather; 8/8 acceptance scenarios; 17/17 browser checks. These are deterministic engineering checks, not a user study or a live-model quality benchmark.

Data provenance

298 named POIs across five Astana neighborhoods, automatically collected from OpenStreetMap on 18 October 2026. 102 have opening hours; 26 have wheelchair tags. Descriptions are explicitly labelled summaries of source category, cuisine and street tags. Source IDs, original tags, collection date and metadata checksum are preserved. OSM-derived data is licensed ODbL 1.0, © OpenStreetMap contributors.

RAG approach

Automated source collection BM25 and multilingual concept retrieval hard geographic/time/context checks recommendations and a short route optional Gemini structured generation validation of source IDs and approved action IDs. Live Gemini structured generation was verified locally and on the private hosted site on 18 October 2026; evidence is recorded in docs/live-generation-check.json. These smoke checks establish connectivity and validated output, not general model quality. Mock tests cover malformed and unsupported responses; provider failures fall back to explicitly labelled templates.

Acceptance scenarios

expo-hour — PASS

Expected: 1–3 nearby options; total route including return 60 minutes.

Observed: 3 recommendations; route estimate 60 / 60 minutes. Places: Лепим и варим, NU Park, Japanese garden.

quiet-coffee — PASS

Expected: Cafés only; quietness and sockets explicitly unverified.

Observed: 3 recommendations; route estimate 28 / 60 minutes. Places: Degirmen, CrepeCafe, Starbucks.

family-museum — PASS

Expected: Family candidate categories; no invented interactive exhibits.

Observed: 2 recommendations; route estimate 104 / 120 minutes. Places: Қазақстан Республикасының Ұлттық Мұражайы, Президент саябағы.

rainy-khan — PASS

Expected: Indoor categories; weather labelled user-selected.

Observed: 3 recommendations; route estimate 90 / 90 minutes. Places: Лепим и варим, Хан Шатыр, Ozyurt.

too-little-time — PASS

Expected: No invented match when nothing fits.

Observed: 0 recommendations; route estimate 0 / 10 minutes. No places returned.

wheelchair — PASS

Expected: Only explicit wheelchair=yes; route access still unverified.

Observed: 2 recommendations; route estimate 44 / 60 minutes. Places: Керуен, Избушка.

airport-night — PASS

Expected: Airport buffer and night safety warnings; no flight guarantee.

Observed: 3 recommendations; route estimate 116 / 120 minutes. Places: Coco, завтрак бар, NU Park.

outside-coverage — PASS

Expected: Empty result outside collected Astana areas.

Observed: 0 recommendations; route estimate 0 / 60 minutes. No places returned.

All non-night scenarios use simulated 19 October 2026, 14:00 Astana (UTC+5). Results are snapshot-based, not claims of verified current availability.

Limits and ethics

Live Gemini structured generation was verified locally and on the private hosted site on 18 October 2026; see `live-generation-check.json`. This verifies connectivity and valid output, not broad model quality. The adapter also has mock tests for invalid responses and falls back to explicitly labelled templates when unavailable.

Generation is deliberately narrow: Gemini chooses an approved activity ID for each retrieved POI. It does not write unrestricted factual prose or extract arbitrary constraints. The app renders localized suggestions. This is constrained retrieval-augmented generation, not a general city chatbot.

OSM data is incomplete and can be stale. Descriptions are transparent tag summaries. Complex hours (holidays, seasonal rules, appointments) are unknown.

Walking estimates use a 1.3 detour factor on straight-line distance at 75 m/min, or 55 m/min in gentle mode. No crossings, barriers, entrances, crowds, weather delays, queues or ticket availability are modeled.

Weather is manual; events, ratings, prices, crowd levels and sockets have no live integration. Unknown opening hours are allowed with a warning unless the user enables the stricter filter.

The interface's structured fields are the reliable constraints. Natural-language parsing recognizes a limited multilingual vocabulary and durations; it is not comprehensive language understanding.

Way centroids can be inaccurate for entrances; the map is context, not navigation.

English, Russian and Kazakh UI copy is implemented; native-speaker review is still desirable.

Precise geolocation is requested only after a user click and browser permission. It lives in page memory; a clear-location action removes it. The app writes no location history, cookies, account database, localStorage, sessionStorage or analytics events. Preset landmarks are available without location permission.

Recommendation ranking and route planning run locally in the browser. Map geometry is included with the app; the browser makes no external map-tile requests. The optional generation request sends only selected public POI IDs, language and three preference booleans (family, quiet, sockets). The server reconstructs trusted records and sends those public records to Gemini. It never forwards the user's precise origin or raw query. Hosting infrastructure and the model provider may still process network metadata. Opening a source link is an explicit external navigation.

POI records are untrusted input. Text is escaped before rendering; map source links are restricted to canonical OSM record URLs. The model is told not to follow source instructions, and its output must contain exactly one allowed action ID for each selected POI. Fabricated IDs, unsupported actions, duplicate records, malformed JSON, HTTP errors and timeouts all trigger the labelled fallback. Venue names, coordinates, hours and accessibility come from source records, never the model.

Unknown opening hours are not “open now.” The parser handles a limited weekly subset and treats unsupported expressions as unknown. Availability is not guaranteed even when a visit fits the published schedule. Accessibility tags are not verified step-free routes. No claims are made about sockets, quietness, child interactivity, night security or lighting without evidence. Activity suggestions are separately labelled.

Local serving is loopback-only. API responses are no-store; bodies are limited to 8 KiB; cross-origin browser writes are rejected. The hosted Site starts private. Before public operation with a paid model key, add authentication/rate limits and budget controls. Keep `.env` ignored and configure hosted secrets

through the hosting provider, never client JavaScript.

Future monetization must not bypass relevance or feasibility filters. Sponsored results must be clearly labelled. Location histories are not part of the business model.

Submission status

Repository: https://github.com/Csadow/qadam-astana (private). Hosted demo: https://qadam-arcsolo-astana.candy-ball-8297.chatgpt.site (owner-private). A source archive is prepared as an alternative. The captioned MP4 is ready, but it has not been uploaded to YouTube. Judge access, YouTube upload and final form submission remain participant actions.